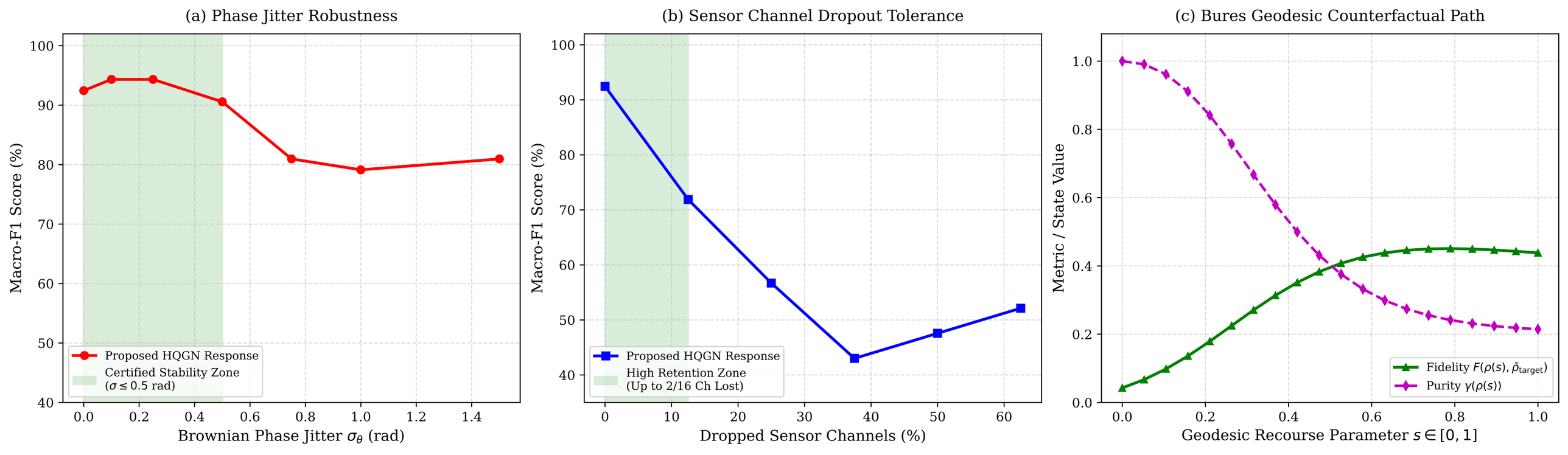
(a) Phase Jitter Robustness
Macro-F1 Score (%)
Brownian Phase Jitter σ_θ (rad)
Proposed HQGN Response
Certified Stability Zone
($\sigma \leq 0.5$ rad)
(b) Sensor Channel Dropout Tolerance
Macro-F1 Score (%)
Dropped Sensor Channels (%)
Proposed HQGN Response
High Retention Zone
(Up to 2/16 Ch Lost)
(c) Bures Geodesic Counterfactual Path
Metric / State Value
Geodesic Recourse Parameter $s \in [0, 1]$
Fidelity $F(\rho(s), \bar{\rho}_{\text{target}})$
Purity $\gamma(\rho(s))$